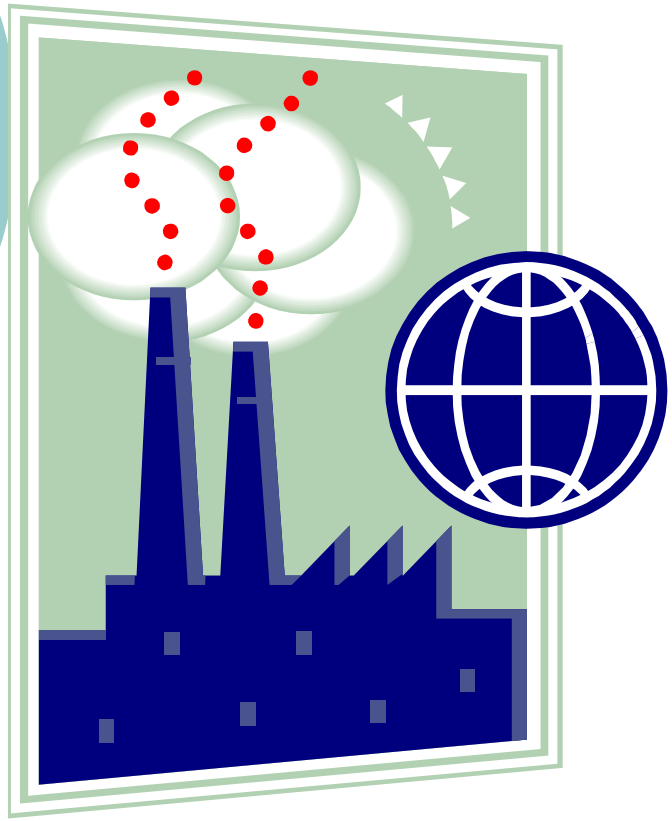
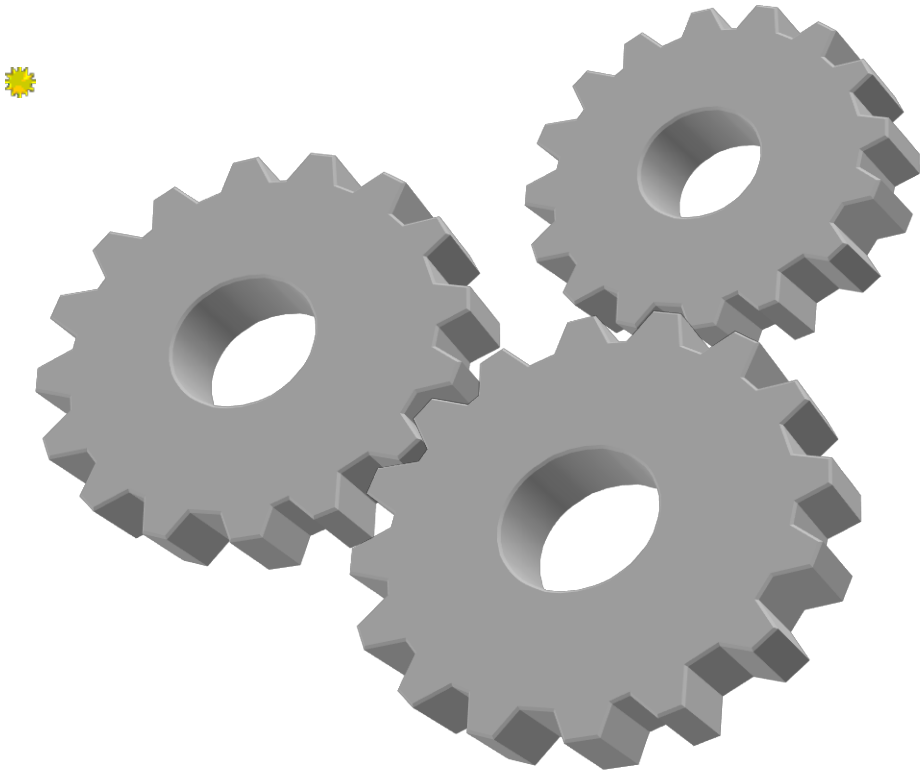
DECISION MAKING





INTRODUCTION

- All managers are decision-makers.
- To carry out the goals of their organizations, managers must have an understanding of how decisions are made and know what decision-making tools are available.
- To a great extent, the success or failure that people and companies experience depends on the quality of their decisions.

THE DECISION PROCESS

- **1. Define the problem and the factors that influence it.**

  - must clear and concise.

- **2. Establish decision criteria and goals.**

  - develop specific and measurable objectives.

- **3. Formulate a model or relationship between goals and variables.**

  - Models contain one or more variables.

  - Variables are measurable quantities that are subject to change.

THE DECISION PROCESS (Contd.)

- **4. Identify and evaluate alternatives.**

  - generate as many solutions to the problem as possible (and usually quickly).

- **5. Select the best alternative.**

  - usually best satisfies and is most consistent with the stated goals.

- **6. Implement the decision.**

  - This stage is sometimes the most challenging (Involves e.g. task assignments and a timetable for implementation)

DECISION THEORY

- Decision theory is an analytic approach to selecting the best alternative or course of action.
- It is used in a wide variety of management situations such as:
  - new product analysis,
  - location planning,
  - equipment selection,
  - scheduling,
  - maintenance planning.

CONDITIONS OF DECISION MAKING

- There are typically three classifications of conditions that decision makers, which depend on the degree of certainty of the possible outcome or the consequences facing the decision maker.
- The three classifications are:
  - Certainty
  - Risk
  - Uncertainty

CERTAINTY

- The *decision-maker knows* with ***certainty*** the consequence or outcome of any alternative or decision choice.

  For example, a decision-maker knows with complete certainty what a Gh₵ 1000 purchase of treasury bills given the stated interest rate and period, will yield upon maturation.

- Decision maker therefore chooses the alternative or option that yields optimal outcome or suits them well.

RISK

- The decision-maker *knows the probability of occurrence* of the outcomes or *consequences* for each choice.
  - *For example, we may not know whether it will rain tomorrow, but we may know that the probability of rain is 0.3.*
- Decision maker under risky conditions seeks to **maximise the expected monetary value.**

UNCERTAINTY

- Decision maker ***does not know*** *the* ***proba*****bility** of occurrence of the outcomes for each alternative.
- The criteria for decision making under uncertainty include
  - **maximax,**
  - **maximin,**
  - **equally likely or Laplace**

MAXIMAX

- This criterion finds an alternative that maximizes the maximum outcome or consequence for every alternative.
- Procedure:
  - First, find the maximum outcome within every alternative
  - Pick the alternative with the maximum number.
  - This decision criterion locates the alternative with the ***highest possible gain***, and hence is called an "**optimistic**" decision criterion.

MAXIMIN

This criterion finds the alternative that maximizes the minimum ***Maximin*** outcome or consequence for every alternative.

- First, find the minimum outcome within every alternative,
- Pick the alternative with the **maximum number**.
- This decision criterion locates the alternative that has the *least* possible loss, it has been called a **"pessimistic**" decision criterion.

EQUALLY LIKELY OR LAPLACE

- This decision criterion finds the alternative with the highest equally likely or **average outcome**.
- Procedure:
  - First calculate the **average outcome** for every alternative (is the sum of all outcomes divided by the number of outcomes).
  - Pick the alternative with the **maximum number**.
  - The equally likely approach assumes that **all probabilities of occurrence for the states of nature are equal** and thus each state of nature is equally likely.

Fundamentals of Decision Theory

- Regardless of the complexity of a decision or the sophistication of the technique used to analyse the decision*, all decision-makers are faced with alternatives and states of nature.*
- In an imperfect and uncertain situation*, all the possible alternatives may not be stated and not all the states of nature will be known*.

Terms

- **Alternative**
  - - a course of action or a strategy that may be chosen by a decision-maker (for example, not carrying an umbrella tomorrow).
- **State of nature**
  - an occurrence or a situation over which the decision-maker has little or no control (for example, tomorrow's weather).

Tools for Decision Making – Decision Trees

Managerial decision making can be represented using **decision trees and decision tables.**

When a ***sequence*** of decisions must be made, **decision trees** are much more powerful tools than **are decision tables**.

Symbols for decision trees:

□ **-**a decision node from which one of several alternatives may be selected.

O - a state of nature node out of which **one** state of nature will occur.

Tools for Decision Making: Decision Tables

- **To develop a decision or payoff table:**
  - List all alternatives down the left side of the table
  - List states of nature (occurrence) across the top of the table
- **For any alternative and a particular state of nature:**
  - there is a consequence or outcome, which is usually expressed as a **monetary value**.
  - This is called a **conditional value**. Conditional values (payoffs) are in the body of the **decision table.**

E.G: Decision For GETZ Products Company

The **Getz Product Company** is investigating the possibility of producing and marketing background storage sheds. With a favourable market, a large facility would give Getz Products a net profit of $ 200, 000. If the market is unfavourable, a $180, 000 net loss would occur. A small plant would result in a net profit of $ 100, 000 in a favourable market, but a net loss of $ 20, 000 would be encountered if the market was unfavourable.

Note: Do nothing is also an alternative

Decision Table for Getz Products

	STATES OF NATURE	
ALTERNATIVES	Favorable Market	Unfavorable Market
Construct large plant	$200,000	-$180,000
Construct small plant	$100,000	-$ 20,000
Do nothing	$0	$0

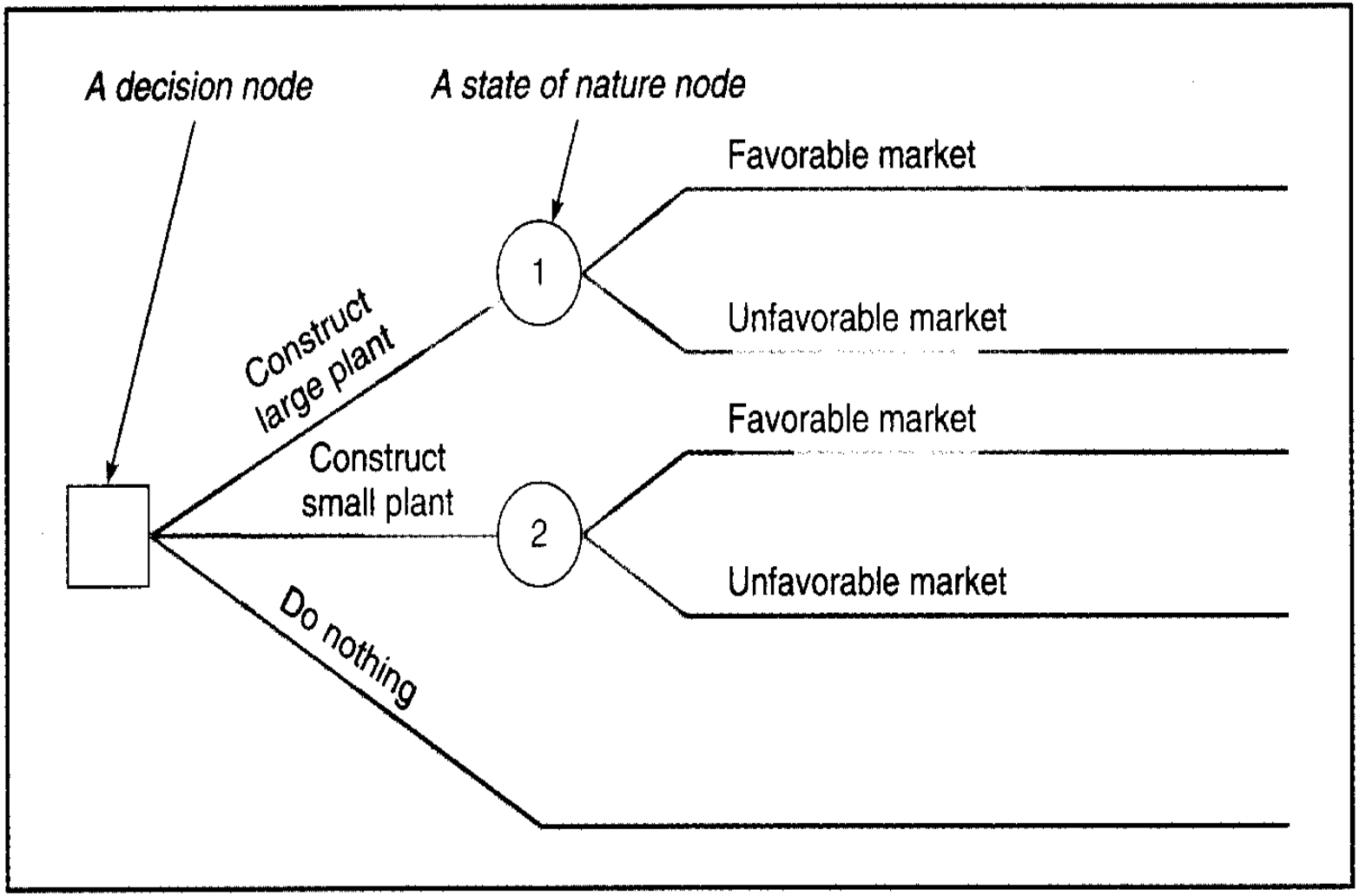


FIGURE S1.1 Getz's decision tree.

Decision Making Under Risk

- Decision making under risk, a more common occurrence, is **a probabilistic decision situation**.
- Several possible states of nature may occur, each with a given probability.
- Given a decision table with conditional values and probability assessments for all states of nature, we can determine the ***expected monetary value (EMV)*** for each alternative.

Expected Monetary Value (EMV)

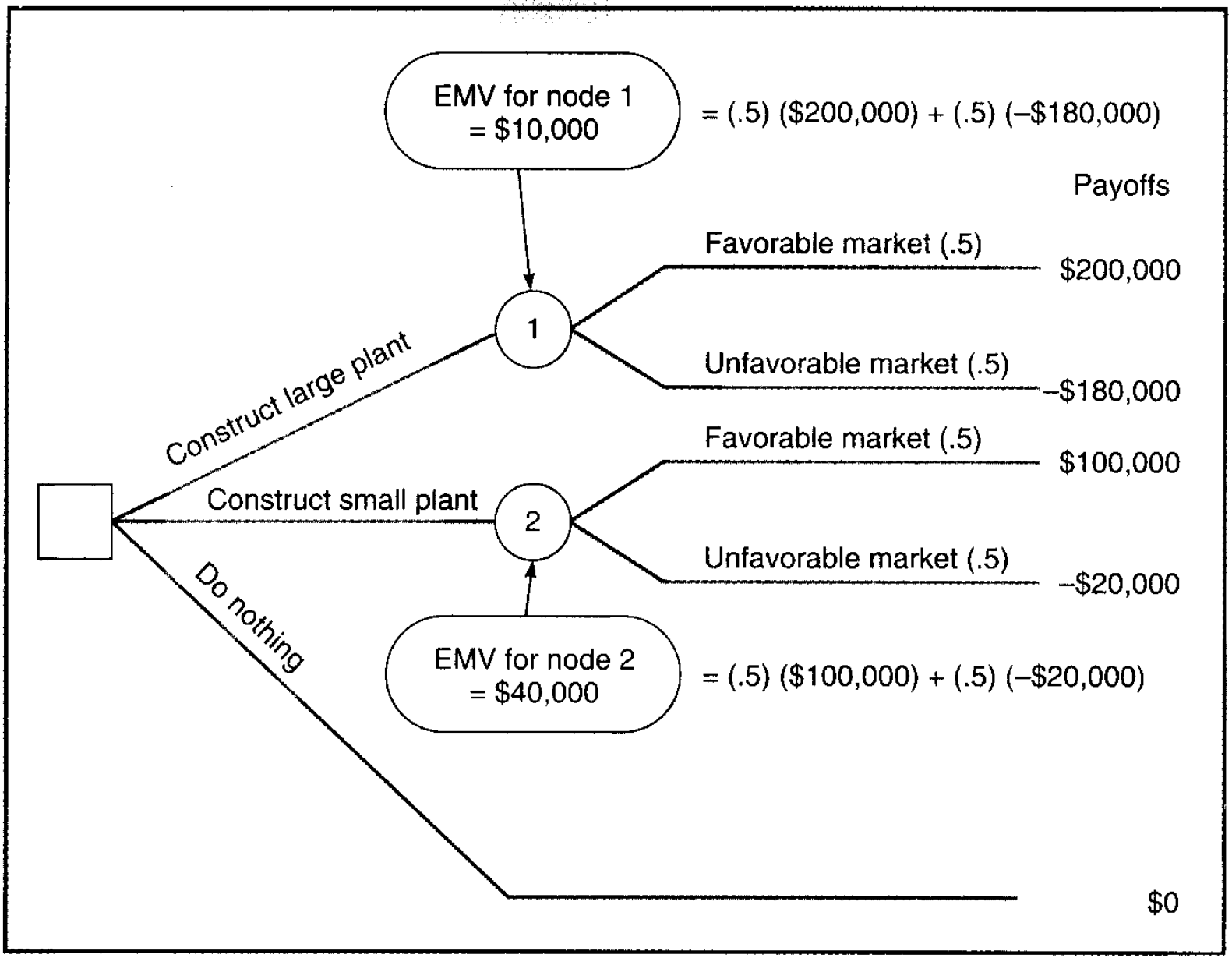
- It is the **average return for each alternative** if we could repeat the decision a large number of times.
- Picking the alternative that has **the maximum EMV** is one of the most popular decision criteria.
- The **EMV** for an alternative is just **the sum of possible payoffs of the alternative, each weighted by the probability** of that payoff occurring.

EMV

- EMV (Alternative 0 = (Payoff of 1 state of nature) x (Probability of 1 state of nature) + (Payoff of 2nd state of nature) x (Probability of 2nd state of nature) + - - - + (Payoff of last state of nature) x (Probability of last state of nature)

EMV CRITERIA

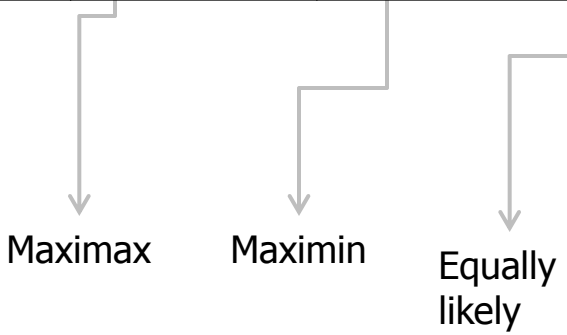
- 1. EMV(A1) = (.5)($200,000) + (.5)(-180,000) = $10,000
- 2. EMV(A2) = (.5)($100,000) + (.5)(-20,000) = $40,000
- 3. EMV(A3) = (.5)($0) + (.5)($0) = $0
- Decision: **build the small facility**.


EMV for node 1
= $10,000
= (.5) ($200,000) + (.5) (–$180,000)
Payoffs
Favorable market (.5)
$200,000
1
Unfavorable market (.5)
–$180,000
Construct large plant
Favorable market (.5)
$100,000
Construct small plant
2
Unfavorable market (.5)
–$20,000
Do nothing
EMV for node 2
= $40,000
= (.5) ($100,000) + (.5) (–$20,000)
$0

DECISIONS UNDER EACH UNCERTAINTY CRITERIA

STATES OF NATURE

ALTERNATIVES	Favourable Market	Unfavourable Market	Maximum in Row	Minimum in Row	Row Average
Construct a large plant	$ 200, 000	($180, 000)	$ 200, 000	($ 180, 000)	$ 10, 000
Construct small plant	$ 100, 000	($20, 000)	$ 100, 000	($ 20, 000)	$ 40, 000
Do nothing	$0	$0	$0	$0	$0



DECISIONS UNDER EACH UNCERTAINTY CRITERIA

- 1 .The maximax choice is to construct a large plant- find the *maximum payoff* and select it

- 2. The maximin choice is to do nothing- find the minimum payoff of each alternative and select *minimum*.

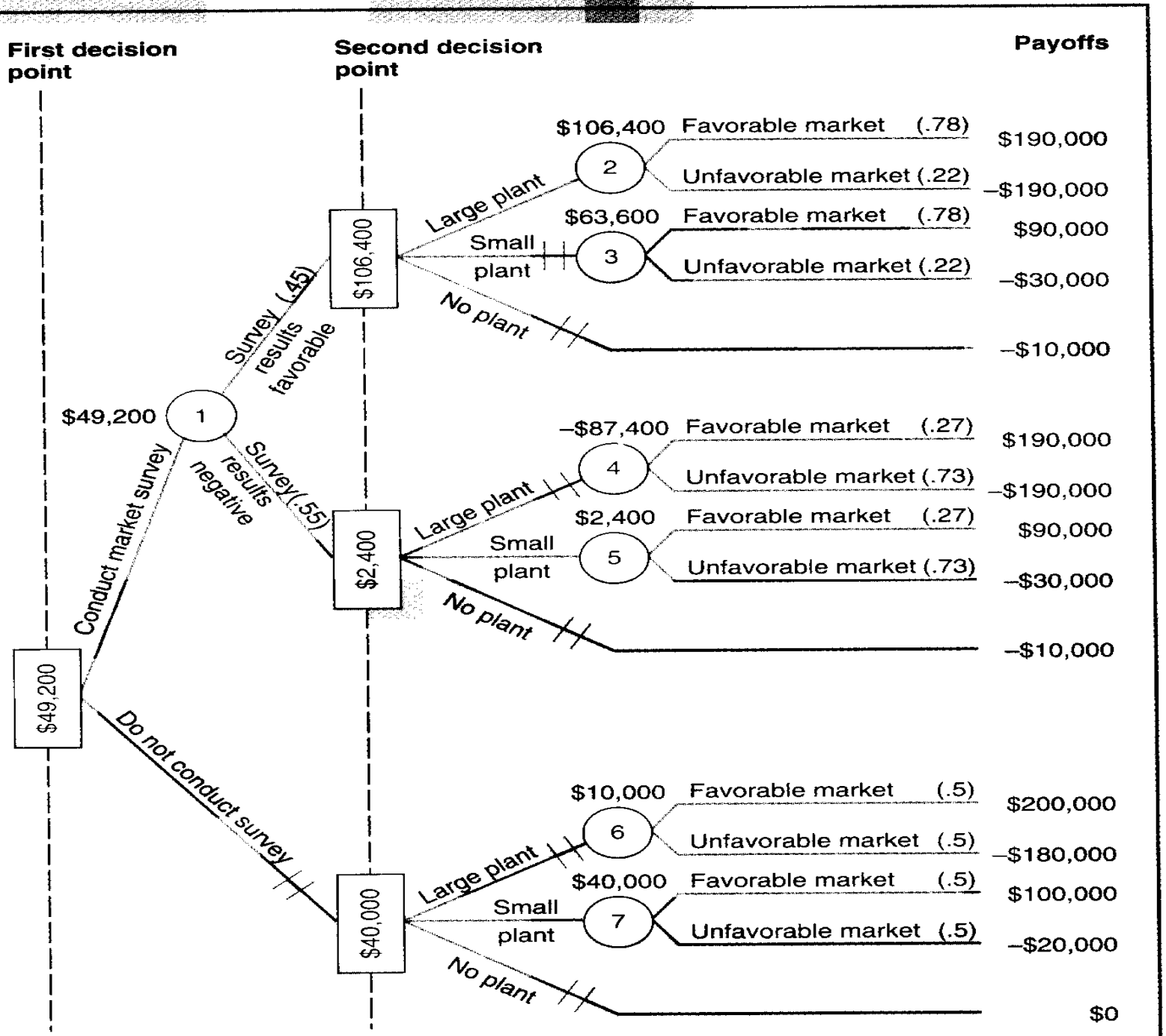
- 3. The equally likely choice is to construct a small plant. This is the maximum of the average outcome of each alternative.

A More Complex Decision Tree

✓Lets assume that GETZ Products has two decisions to make, with the second one dependent on the first.

✓Before deciding on whether to build a new plant, GETZ Products has an option to conduct a marketing research survey, at a cost of $ 10, 000.

✓The information from the survey could help decide whether to build a large plant, a small plant or not to build at all.


First decision point
Second decision point
Payoffs
$106,400
Favorable market (.78)
$190,000
2
Unfavorable market (.22)
–$190,000
Large plant
$63,600
Favorable market (.78)
$90,000
Small plant
3
Unfavorable market (.22)
–$30,000
No plant
–$10,000
$106,400
Survey (.45) results favorable
$49,200
1
–$87,400
Favorable market (.27)
$190,000
4
Unfavorable market (.73)
–$190,000
Large plant
$2,400
Favorable market (.27)
$90,000
Small plant
5
Unfavorable market (.73)
–$30,000
No plant
–$10,000
$2,400
Survey (.55) results negative
Conduct market survey
$49,200
Do not conduct survey
$10,000
Favorable market (.5)
$200,000
6
Unfavorable market (.5)
–$180,000
Large plant
$40,000
Favorable market (.5)
$100,000
Small plant
7
Unfavorable market (.5)
–$20,000
No plant
$0
$40,000

USES OF DECISION ANALYSIS

- Decision trees and tables are especially useful for making decisions under risk and uncertainty.
- Investments in research and development, plant and equipment, and even new buildings and structures can be analyzed with decision theory.
- Problems in inventory control, aggregate planning, maintenance, scheduling, and production control are just a few other decision theory applications.